28/01/2025
Morning

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Medical | IIT-JEE | Foundations

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Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2025 Phase-1
[Computer Based Test (CBT) mode]

(Mathematics, Physics and Chemistry)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 75 questions. Each subject (MPC) has 25 questions. The maximum marks are 300.

(3) This question paper contains **Three** Parts. **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is **Mathematics**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt all questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 25) contains** 5 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. Let $\langle a_n \rangle$ be a sequence such that $a_0 = 0$, $a_1 = \frac{1}{2}$ and $2a_{n+2} = 5a_{n+1} - 3a_n$, $n = 0, 1, 2, 3, \ldots$. Then $\sum_{k=1}^{100} a_k$ is equal to

(1) $3a_{100} - 100$ (2) $3a_{100} + 100$

(3) $3a_{99} - 100$ (4) $3a_{99} + 100$

Answer (1)

Sol. $2a_{n+2} = 5a_{n+1} - 3a_n$, $n = 0, 1, 2, \cdots$ using characteristic equation

$2x^2 = 5x - 3$

$\Rightarrow \quad x = 1, \frac{3}{2}$

$\Rightarrow \quad a_n = p(1)^n + q\left(\frac{3}{2}\right)^n$

$\Rightarrow \quad a_0 = p + q = 0$

$a_1 = p + \frac{3}{2}q = \frac{1}{2} \Rightarrow q = 1, \; p = -1$

$\Rightarrow \quad a_n = \left(\frac{3}{2}\right)^n - 1$

$\sum_{k=1}^{100} a_k = \sum_{k=1}^{100}\left(\frac{3}{2}\right)^k - \sum_{k=1}^{100}(1)$

$= \frac{\left(\frac{3}{2}\right)\left[\left(\frac{3}{2}\right)^{100} - 1\right]}{\left(\frac{3}{2} - 1\right)} - 100$

$= 3\left[\left(\frac{3}{2}\right)^{100} - 1\right] - 100$

$= 3a_{100} - 100$

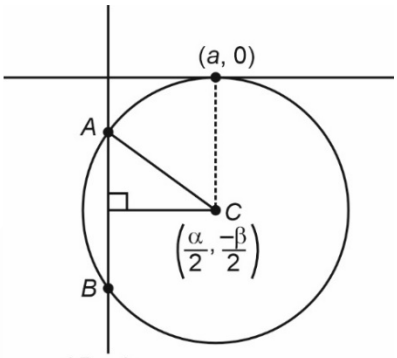
2. Let the equation of the circle, which touches x-axis at the point $(a, 0)$, $a > 0$ and cuts off an intercept of length b on y-axis be $x^2 + y^2 - \alpha x + \beta y + \gamma = 0$. If the circle lies below x-axis, then the ordered pair $(2a, b^2)$ is equal to

(1) $(\alpha, \beta^2 + 4\gamma)$ (2) $(\gamma, \beta^2 - 4\alpha)$

(3) $(\gamma, \beta^2 + 4\alpha)$ (4) $(\alpha, \beta^2 - 4\gamma)$

Answer (4)

Sol.



$AB = b$

$r = \left|\frac{-\beta}{2}\right|$

$a = \frac{\alpha}{2} \Rightarrow 2a = \alpha$

$\Rightarrow \quad r^2 = \left(\frac{b}{2}\right)^2 + \left(\frac{\alpha}{2}\right)^2$

$\Rightarrow \quad \frac{\beta^2}{4} = \frac{b^2}{4} + a^2$

$\Rightarrow \quad b^2 = \beta^2 - 4a^2 = \beta^2 - (2a)^2 = \beta^2 - \alpha^2$

$r^2 = \frac{\beta^2}{4} = \frac{\alpha^2}{4} + \frac{\beta^2}{4} - \gamma \Rightarrow \gamma = \frac{\alpha^2}{4}$

$\Rightarrow \quad b^2 = \beta^2 - 4\gamma$

$(2a, b^2) = (\alpha, \beta^2 - 4\gamma)$

3. Two number k_1 and k_2 are randomly chosen from the set of natural numbers. Then, the probability that the value of $i^{k_1} + i^{k_2}$, $(i = \sqrt{-1})$ is non-zero, equals

(1) $\frac{2}{3}$ (2) $\frac{1}{4}$

(3) $\frac{3}{4}$ (4) $\frac{1}{2}$

Answer (3)



Sol. Let $k_1 = 4\lambda_1 + r_1$, $r_i \in \{0, 1, 2, 3\}$

$k_2 = 4\lambda_2 + r_2$

$(i)^{k_1} + (i)^{k_2} = (i)^{r_1} + (i)^{r_2}$

$(i)^{r_1} \in \{1, i, -1, -i\}$

Zero $\Rightarrow$ 1, (–1) pair $\Rightarrow \begin{Bmatrix} 1, & -1 \\ i, & -i \\ -i, & +i \\ -1, & 1 \end{Bmatrix}$

i, ($-i$) pair

Zero probability = $\dfrac{4}{{}^4C_1 \cdot {}^4C_1} = \dfrac{1}{4}$

Probability (non-zero) $= 1 - \dfrac{1}{4} = \dfrac{3}{4}$

4. The number of different 5 digit numbers greater than 50000 that can be formed using the digits 0, 1, 2, 3, 4, 5, 6, 7, such that the sum of their first and last digits should not be more than 8, is

(1) 5719 (2) 4608

(3) 5720 (4) 4607

Answer (4)

Sol. $\underline{a}\ \underline{b}\ \underline{c}\ \underline{d}\ \underline{e}$

$a \geq 5$

all b, c, d, e are not 0

(i) $a = 5$

$5\ _\ _\ _\ \underline{e} \Rightarrow e \in \{3, 2, 1, 0\}$

$8^3 \cdot 4 - 1$ ways

(ii) $a = 6$

$6\ _\ _\ _\ \underline{e} \Rightarrow e \in \{2, 1, 0\}$

$8^3 \cdot 3$ ways

(iii) $7\ _\ _\ _\ \underline{e} \Rightarrow e \in \{1, 0\}$

$8^3 \cdot 2$ ways

Total = $8^3[4 + 3 + 2] - 1$

= 4607 numbers

5. Let ${}^nC_{r-1} = 28$, ${}^nC_r = 56$ and ${}^nC_{r+1} = 70$. Let A (4cos$t$, 4sin$t$), B (2sin$t$, –2cos$t$) and C (3r–n, $r^2 - n - 1$) be the vertices of a triangle ABC, where t is a parameter. If $(3x - 1)^2 + (3y)^2 = \alpha$, is the locus of the centroid of triangles ABC, then α equals

(1) 6 (2) 18

(3) 8 (4) 20

Answer (4)

Sol. ${}^nC_{r-1} = 28$

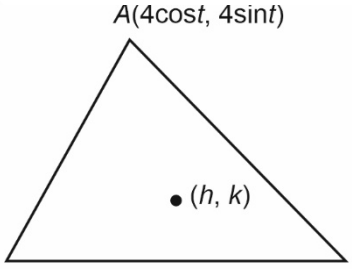
${}^nC_r = 56$

${}^nC_{r+1} = 70$

$\dfrac{{}^nC_{r-1}}{{}^nC_r} = \dfrac{28}{56} \Rightarrow \dfrac{r}{n - r + 1} = \dfrac{1}{2}$

$\dfrac{{}^nC_r}{{}^nC_{r+1}} = \dfrac{56}{70} \Rightarrow \dfrac{r + 1}{n - r} = \dfrac{70}{56}$

$\Rightarrow n = 8$, $r = 3$



$h = \dfrac{4\cos t + 2\sin t + 1}{3}$ $\quad k = \dfrac{4\sin t - 2\cos t}{3}$

$3h - 1 = 4\cos t + 2\sin t$ …(1)

$3k - 1 = 4\sin t - 2\cos t$ …(2)

$(1)^2 + (2)^2$

$(3h - 1)^2 + (3k)^2 = 20$

Locus of centroid: $(3x - 1)^2 + (3y)^2 = 20$

$\Rightarrow \alpha = 20$

6. Let T_r be the rth term of an A.P. If for some m, $T_m = \dfrac{1}{25}$, $T_{25} = \dfrac{1}{20}$, and $20\sum_{r=1}^{25} T_r = 13$, then $5m\sum_{r=m}^{2m} T_r$ is equal to

(1) 142 (2) 126

(3) 112 (4) 98

Answer (2)



Sol. $20\sum_{r=1}^{25} T_r = 20\left[\frac{25}{2}\left[a+\frac{1}{20}\right]\right] = 13$

$\Rightarrow \quad a = \frac{1}{20\times 25}$

$\because \quad T_{25} = a + 24d = \frac{1}{20}$

$\Rightarrow \quad d = \frac{1}{20\times 25}$

$T_m = a + (m-1)d = \frac{1}{25}$

$= \frac{1}{20\times 25}m = \frac{1}{25} \Rightarrow \boxed{m = 20}$

Now, $5m\sum_{r=m}^{2m} T_r = 5\times 20\left[\sum_{r=20}^{40} T_r\right]$

$= 100\left[\frac{40}{2}(2a+39d) - \frac{19}{2}(2a+18d)\right]$

$\because \quad a = d$

$\Rightarrow \quad 100\left[\frac{40}{2}\times 41d - \frac{19}{2}\times 21d\right]$

$= 126$

7. Let O be the origin, the point A be $z_1 = \sqrt{3}+2\sqrt{2}i$, the point $B(z_2)$ be such that $\sqrt{3}|z_2| = |z_1|$ and $\arg(z_2) = \arg(z_1) + \frac{\pi}{6}$. Then

(1) ABO is a scalene triangle

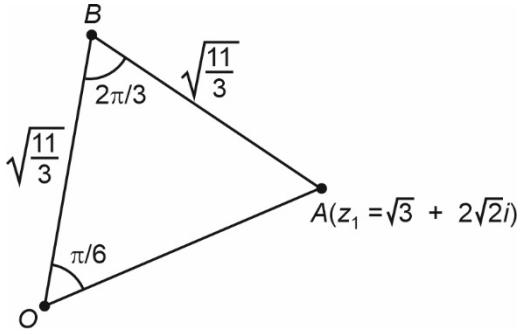
(2) ABO is an obtuse angled isosceles triangle

(3) area of triangle ABO is $\frac{11}{\sqrt{3}}$

(4) area of triangle ABO is $\frac{11}{4}$

Answer (2)

Sol.



$OA = |z_1| = \sqrt{3+8} = \sqrt{11}$

and $OB = \frac{1}{\sqrt{3}}|z_1| = \sqrt{\frac{11}{3}}$

$AB^2 = OA^2 + OB^2 - 2\cdot OA\cdot OB\cos\frac{\pi}{6}$

$= 11 + \frac{11}{3} - 2\cdot\frac{11}{\sqrt{3}}\cdot\frac{\sqrt{3}}{2}$

$\therefore \quad AB = \sqrt{\frac{11}{3}}$

$\therefore$ Area of $\Delta ABD = \frac{1}{2}\cdot OA\cdot OB\cdot\sin\frac{\pi}{6}$

$= \frac{11}{4\sqrt{3}}$ sq. units

Here $OB = AB$ and $\angle A = \frac{2\pi}{3}$

$\therefore \quad \Delta ABD$ is an obtuse angled isosceles triangle.

8. Three defective oranges are accidently mixed with seven good ones and on looking at them, it is not possible to differentiate between them. Two oranges are drawn at random from the lot. If x denotes the number of defective oranges, then the variance of x is

(1) 18/25 (2) 26/75

(3) 28/75 (4) 14/25

Answer (3)

Sol. There are 3 bad oranges and 7 good oranges.

$\therefore \quad X$ = number of bad oranges drawn.

X	0	1	2
$P(X)$	$\frac{{}^7C_2}{{}^{10}C_2}$	$\frac{{}^3C_1\times{}^7C_1}{{}^{10}C_2}$	$\frac{{}^3C_2}{{}^{10}C_2}$



$\therefore$ Variance

$$= 0^2 \cdot \frac{{}^7C_2}{{}^{10}C_2} + 1^2 \cdot \left(\frac{3\times 7}{{}^{10}C_2}\right) + 2^2\left(\frac{3}{{}^{10}C_2}\right) - \left(0 + 1\cdot\frac{3\times 7}{{}^{10}C_2} + 2\cdot\frac{3}{{}^{10}C_2}\right)^2$$

$$= \frac{28}{75}$$

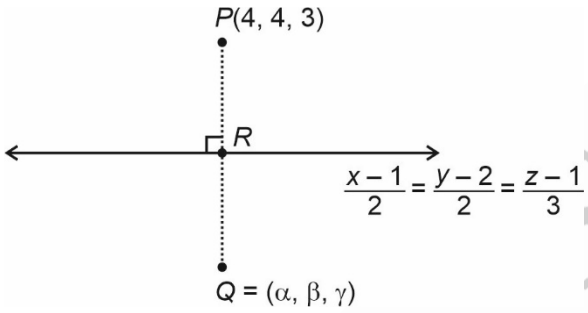
9. If the image of the point (4, 4, 3) in the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-1}{3}$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to

(1) 12 (2) 7

(3) 9 (4) 8

Answer (3)

Sol.



Let coordinate of $R = (2r + 1, r + 2, 3r + 1)$

$\because$ PR is perpendicular to given line.

$\therefore$ $(2r - 3)\cdot 2 + (r - 2)\cdot 1 + (3r - 2)\cdot 3 = 0$

$\therefore$ $r = 1$

$\therefore$ Coordinate of $R = (3, 3, 4)$

$\therefore$ $(\alpha, \beta, \gamma) = (2, 2, 5)$

$\therefore$ $\alpha + \beta + \gamma = 9$

10. $\cos\left(\sin^{-1}\frac{3}{5} + \sin^{-1}\frac{5}{13} + \sin^{-1}\frac{33}{65}\right)$ is equal to:

(1) 1 (2) $\frac{33}{65}$

(3) $\frac{32}{65}$ (4) 0

Answer (4)

Sol. $\cos\left(\sin^{-1}\frac{3}{5} + \sin^{-1}\left[\frac{5}{13}\sqrt{1-\frac{33^2}{65^2}} + \frac{33}{65}\sqrt{1-\frac{5^2}{13^2}}\right]\right)$

$$= \cos\left(\sin^{-1}\frac{3}{5} + \sin^{-1}\frac{4}{5}\right)$$

$$= \cos\left(\sin^{-1}\frac{3}{5} + \cos^{-1}\frac{3}{5}\right)$$

$$= \cos\left(\frac{\pi}{2}\right) = 0$$

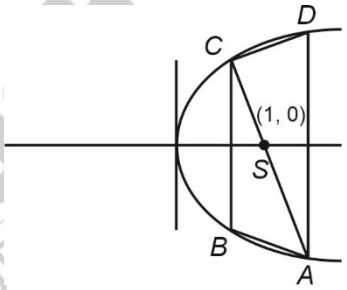
11. Let $ABCD$ be a trapezium whose vertices lie on the parabola $y^2 = 4x$. Let the sides AD and BC of the trapezium be parallel to y-axis. If the diagonal AC is of length $\frac{25}{4}$ and it passes through the point (1, 0), then the area of $ABCD$ is

(1) $\frac{125}{8}$ (2) $\frac{25}{2}$

(3) $\frac{75}{8}$ (4) $\frac{75}{4}$

Answer (4)

Sol.



Take $AS = c$

$$\frac{1}{AS} + \frac{1}{CS} = \frac{1}{a}$$

$$\Rightarrow \frac{1}{c} + \frac{1}{\frac{25}{4} - c} = \frac{1}{a}$$

$$\Rightarrow 4c^2 - 25c + 25 = 0$$

$$\Rightarrow c = \frac{5}{4}, 5$$

$AS = c = 1 + t^2$

$c = \frac{5}{4} \Rightarrow t = \pm\frac{1}{2}$

$c = 5 \Rightarrow t = \pm 2$



$(at^2, 2at) = A\left(\frac{1}{4}, 1\right), D\left(\frac{1}{4}, -1\right), B(4, 4)$

$C(4, -4) \Rightarrow AD = 2,\ BC = 8$

Area of trapezium $= \frac{1}{2}(2+8)\times\frac{15}{4}$

$= \frac{75}{4}$

12. If $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{96x^2\cos^2 x}{(1+e^x)}dx = \pi(\alpha\pi^2+\beta), \alpha,\beta \in \mathbb{Z},$

then $(\alpha + \beta)^2$ equals

(1) 100 (2) 196

(3) 144 (4) 64

Answer (1)

Sol. $I = \int_0^{\frac{\pi}{2}} \frac{96x^2\cos^2 x}{1+e^x}dx$

$2I = 2\int_0^{\frac{\pi}{2}} 96x^2\cos^2 x dx$

$I = 96\int_0^{\frac{\pi}{2}} x^2\cos^2 x dx$

$= 48\int_0^{\frac{\pi}{2}} x^2(1+\cos 2x)dx$

$= 2\pi^2 + 48(0-0) - 48\int_0^{\frac{\pi}{2}} x\sin 2x dx$

$= 2\pi^2 - 12\pi + [0-0] = \pi(2\pi^2 - 12)$

$= \pi(\alpha\pi^2 + \beta)$

$\Rightarrow (\alpha + \beta)^2 = 100$

13. The sum of all local minimum values of the function

$$f(x) = \begin{cases} 1-2x, & x < -1 \\ \frac{1}{3}(7+2|x|), & -1 \le x \le 2 \\ \frac{11}{18}(x-4)(x-5), & x > 2 \end{cases}$$ is

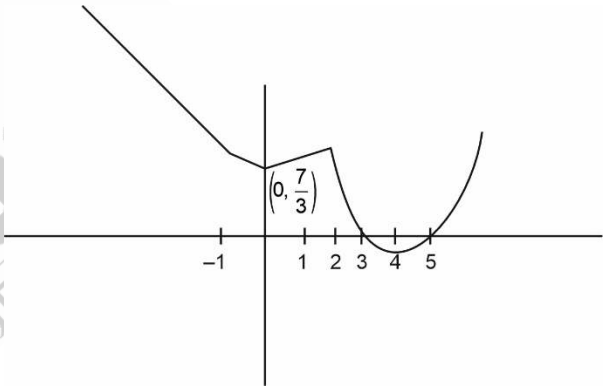
(1) $\frac{131}{72}$ (2) $\frac{167}{72}$

(3) $\frac{171}{72}$ (4) $\frac{157}{72}$

Answer (4)

Sol. $f(x) = \begin{cases} 1-2x, & x < -1 \\ \frac{1}{3}(7+2|x|), & -1 \le x \le 2 \\ \frac{11}{18}(x-4)(x-5), & x > 2 \end{cases}$

$$\Rightarrow f(x) = \begin{cases} 1-2x, & x < -1 \\ \frac{1}{3}(7-2x, & -1 \le x < 0 \\ \frac{1}{3}(7+2x), & 0 \le x \le 2 \\ \frac{11}{8}(x-4)(x-5), & x > 2 \end{cases}$$



Minimum value at $x = 0$ and $x = 4.5$

at $x = 0$, $f(0) = \frac{7}{3}$

at $x = 4.5$, $f(4.5) = \frac{-11}{72}$

$\Rightarrow$ Total sum $= \frac{7}{3} - \frac{11}{72}$

$= \frac{157}{72}$

$\Rightarrow$ Option (4) is correct



14. Let $A(x, y, z)$ be a point in xy-plane, which is equidistant from the points (0, 3, 2), (2, 0, 3) and (0,0,1).

Let $B = (1, 4, -1)$ and $C = (2, 0, -2)$. Then among the statements

($S1$) : ΔABC is an isosceles right angled triangle, and

($S2$) : The area of ΔABC is $\frac{9\sqrt{2}}{2}$,

(1) Both are true (2) Only ($S1$) is true

(3) Only ($S2$) is true (4) Both are false

Answer (2)

Sol. $A(x, y, z)$ $P(0, 3, 2)$, $Q(2, 0, 3)$, $R(0, 0. 1)$

$\Downarrow$

As A is in xy-plane

$\Rightarrow$ $A(x, y, 0)$

$AP^2 = AQ^2$

$(x-0)^2 + (y-3)^2 + (0-2)^2 = (x-2)^2 + (y-0)^2 + (0-3)^2$

$x^2 + y^2 - 6y + 9 + 4 = x^2 - 4x + 4 + y^2 + 9$

$\Rightarrow -6y = -4x$

$\Rightarrow 2x = 3y$ … (i)

$AP^2 = AR^2$

$x^2 + y^2 - 6y + 9 + 4 = x^2 + y^2 + 1$

$\Rightarrow y = 2$

$\Rightarrow x = 3$

$A(3, 2, 0)$, $B(1, 4, -1)$, $C(2, 0, -2)$

$\overrightarrow{AB} = -2\hat{i} + 2\hat{j} - \hat{k}$

$\overrightarrow{BC} = \hat{i} - 4\hat{j} - \hat{k}$

$\overrightarrow{AC} = -\hat{i} - 2\hat{j} - 2\hat{k}$

$\overrightarrow{AB} \cdot \overrightarrow{AC} = 2 - 4 + 2 = 0$

$\Rightarrow \Delta ABC$ is right angle triangle at A

Also, $AB = 3$, $BC = 3\sqrt{2}$, $AC = 3$

$\Rightarrow AB = AC$

$\text{ar}(\Delta ABC) = \frac{1}{2} \times AB \times AC = \frac{9}{2}$

$\Rightarrow$ only ($S1$) is true

$\Rightarrow$ option (2) is correct

15. The relation $R = \{(x, y) : x, y \in \mathbb{Z}$ and $x + y$ is even$\}$ is:

(1) Reflexive and transitive but not symmetric

(2) Symmetric and transitive but not reflexive

(3) Reflexive and symmetric but not transitive

(4) An equivalence relation

Answer (4)

Sol. For reflexive $(x, x) \in \mathbb{R}, x \in \mathbb{Z}$

$\Rightarrow x + x = 2x \rightarrow$ even

For symmetric of $(x, y) \in \mathbb{R}$ then $(y, x) \in \mathbb{R}$

when $x, y \in \mathbb{Z}$

$x + y \rightarrow$ even

$\Rightarrow y + x \rightarrow$ even

for transitive if $(x, y) \in \mathbb{R} \Rightarrow x + y \rightarrow$ even

$(y, z) \in \mathbb{R} \Rightarrow y + z \rightarrow$ even

$x + 2y + z \rightarrow$ even

$\Rightarrow x + z$ is even

$\Rightarrow (x, z) \in \mathbb{R}$

$\Rightarrow \mathbb{R}$ is an equivalence relation.

$\Rightarrow$ Option (4) is correct.

16. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$f(x) = (2+3a)x^2 + \left(\frac{a+2}{a-1}\right)x + b, a \neq 1$. If $f(x + y) = f(x) + f(y) + 1 - \frac{2}{7}xy$, then the value of $28\sum_{i=1}^{5}|f(i)|$ is

(1) 715 (2) 675

(3) 735 (4) 545

Answer (2)

Sol. Put $y = 0$

$f(x) = f(0) + f(x) + 1 - 0$

$f(0) = -1$

$f(0) = 0 + 0 + b$

$\Rightarrow b = -1$

$f(-1 + 1) = f(-1) + f(1) + 1 + \frac{2}{7}$

$f(0) = f(-1) + f(1) + \frac{9}{7}$



$$-1=(2+3a)+\left(\frac{a+2}{a-1}\right)(-1)+b+(2+3a)$$

$$+\frac{a+2}{a-1}+b+\frac{9}{7}$$

$$-1=4+6a-2+\frac{9}{7}$$

$$-1=2+\frac{9}{7}+6a$$

$$6a=-1-2-\frac{9}{7}$$

$$a=\frac{-5}{7}$$

$$f(x)=\frac{-x^2}{7}+\frac{\frac{9}{7}}{\frac{-12}{7}}x-1$$

$$f(x)=\frac{-x^2}{7}-\frac{3}{4}x-1$$

$$\sum_{i=1}^{5}f(i)=-\frac{1}{7}\left(\frac{5\times6\times11}{6}\right)-\frac{3}{4}\left(\frac{5\times6}{2}\right)-5$$

$$=\frac{-55}{7}-\frac{45}{4}-5$$

$$=\frac{675}{28}$$

$$\Rightarrow 28\left|\sum_{i=1}^{5}f(i)\right|=675$$

17. The area (in sq. units) of the region

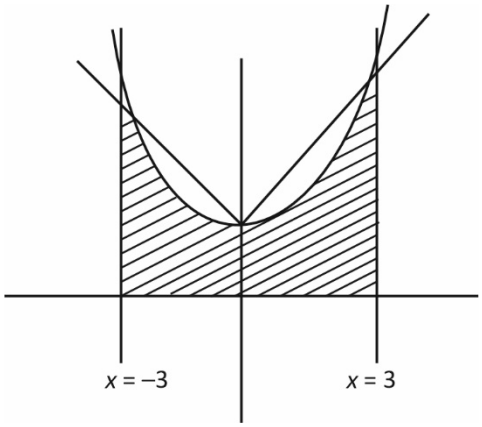
$\{(x, y) : 0 \le y \le 2|x| + 1, 0 \le y \le x^2 + 1, |x| \le 3\}$ is

(1) $\frac{17}{3}$ (2) $\frac{80}{3}$

(3) $\frac{64}{3}$ (4) $\frac{32}{3}$

Answer (3)

Sol.



$$\text{Area}=2\left[\int_0^2(x^2+1)dx+\frac{1}{2}[5+7]\times1\right]$$

$$=\frac{64}{3}$$

18. If $f(x)=\frac{2^x}{2^x+\sqrt{2}}, x\in\mathbb{R}$, then

$\sum_{k=1}^{81}f\left(\frac{k}{82}\right)$ is equal to

(1) $81\sqrt{2}$ (2) 82

(3) $\frac{81}{2}$ (4) 41

Answer (3)

Sol. $f(x)=\frac{2^x}{2^x+2^{1/2}}=\frac{2^x}{2^x+\sqrt{2}}$

$$f(1-x)=\frac{2^{1-x}}{2^{1-x}+2^{1/2}}=\frac{\frac{2}{2^x}}{\frac{2}{2^x}+2^{1/2}}=\frac{2}{2+\sqrt{2}\,2^x}$$

$$=\frac{\sqrt{2}}{2^x+\sqrt{2}}$$

$$\Rightarrow\quad f(x)+f(1-x)=\frac{\sqrt{2}+2^x}{\sqrt{2}+2^x}=1$$

$$\Rightarrow\quad \sum_{k=1}^{81}f\left(\frac{k}{82}\right)+f\left(\frac{2}{82}\right)+\left(f\left(\frac{3}{82}\right)\right)+....$$

$$...+f\left(\frac{40}{82}\right)+f\left(\frac{41}{82}\right)+f\left(\frac{42}{82}\right)$$

$$+...+f\left(\frac{79}{82}\right)+f\left(\frac{80}{82}\right)+f\left(\frac{81}{82}\right)$$



$$=\left[f\left(\frac{1}{82}\right)+f\left(\frac{81}{82}\right)\right]+\left[f\left(\frac{2}{82}\right)+f\left(\frac{80}{82}\right)\right]+\ldots$$
$$+\left[f\left(\frac{40}{82}\right)+f\left(\frac{42}{82}\right)+f\left(\frac{41}{82}\right)\right]$$
$$=\underbrace{1+1+\ldots.+1}_{40\text{ times}}+f\left(\frac{1}{2}\right)$$
$$=40+\frac{\sqrt{2}}{\sqrt{2}+\sqrt{2}}=40+\frac{1}{2}=\frac{81}{2}$$

19. Let for some function $y=f(x)$, $\int_0^x tf(t)dt = x^2f(x), x>0$ and $f(2)=3$. Then $f(6)$ is equal to

(1) 6 (2) 3
(3) 1 (4) 2

Answer (3)

Sol. $\int_0^x tf(t)dt = x^2f(x)$

$\Rightarrow xf(x) = 2xf(x) + x^2 f'(x)$

$\Rightarrow x^2.f'(x) = -x f(x)$

$\Rightarrow \frac{f'(x)}{f(x)} = -\frac{1}{x}$

$\int \frac{f'(x)}{f(x)dx} = -\int \frac{1}{x}dx$

$\Rightarrow \ln f(x) = -\ln x + C$

$f(2) = 3 \Rightarrow \ln 3 = -\ln 2 + C$

$C = \ln 6$

$\Rightarrow f(x) = \frac{6}{x} \Rightarrow f(6) = 1$

20. The sum, of the squares of all the roots of the equation $x^2 + |2x-3| - 4 = 0$, is

(1) $3(2-\sqrt{2})$ (2) $6(3-\sqrt{2})$
(3) $3(3-\sqrt{2})$ (4) $6(2-\sqrt{2})$

Answer (4)

Sol. For $x \geq \frac{3}{2}$

$x^2 + 2x - 3 - 4 = 0$

$x^2 + 2x - 7 = 0$

$x = \frac{-2 \pm \sqrt{4+28}}{2} = -1 \pm 2\sqrt{2}$

Only $2\sqrt{2}-1$ is acceptable root

For $x < \frac{3}{2}$ $\quad x^2 - 2x + 3 - 4 = 0$

$x^2 - 2x - 1 = 0$

$x = \frac{2 \pm \sqrt{4+4}}{2} = 1 \pm \sqrt{2}$

Only $1-\sqrt{2}$ is acceptable

Sum of the square = $\left(1-\sqrt{2}\right)^2 + \left(2\sqrt{2}-1\right)^2$

$= 6\left(2-\sqrt{2}\right)$

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. If $\alpha = 1 + \sum_{r=1}^{6} (-3)^{r-1}\ {}^{12}C_{2r-1}$, then the distance of the point $(12, \sqrt{3})$ from the line $\alpha x - \sqrt{3}y + 1 = 0$ is _____.

Answer (5)

Sol. $\alpha = 1 + \sum_{r=1}^{6} (-1)^{r-1}\ {}^{12}C_{2r-1}3^{r-1}$

Let $\sqrt{3}i = x$

$\alpha = 1 + \sum_{r=1}^{6} \left[(\sqrt{3}i)^2\right]^{r-1} \cdot {}^{12}C_{2r-1}$

$\alpha = 1 + \sum_{r=1}^{6} {}^{12}C_{r-1} \frac{(\sqrt{3}i)^{2r-1}}{\sqrt{3}i}$

$\alpha = 1 + \frac{1}{\sqrt{3}i}\left({}^{12}C_1 x + {}^{12}C_3 x^3 + \ldots {}^{12}C_{11}x^{11}\right)$

$= 1 + \frac{1}{\sqrt{3}i}\left(\frac{(1+\sqrt{3}i)^{12} - (1-\sqrt{3}i)^{12}}{2}\right)$

$= 1 + \frac{1}{\sqrt{3}i}\left(\frac{(-2\omega^2)^{12} - (2\omega)^{12}}{2}\right) = 1$

So, distance of $(12, \sqrt{3})$ from $x - \sqrt{3}y + 1 = 0$ is

$\frac{12 - 3 + 1}{2} = 5$



22. Let M denote the set of all real matrices of order 3×3 and let $S = \{-3, -2, -1, 1, 2\}$. Let

$S_1 = \{A = [a_{ij}] \in M : A = A^T \text{ and } a_{ij} \in S, \forall\, i, j\}$,

$S_2 = \{A = [a_{ij}] \in M : A = -A^T \text{ and } a_{ij} \in S, \forall\, i, j\}$,

$S_3 = \{A = [a_{ij}] \in M : a_{11} + a_{22} + a_{33} = 0 \text{ and } a_{ij} \in S, \forall\, i, j\}$.

If $n(S_1 \cup S_2 \cup S_3) = 125\,\alpha$, then α equals _____.

Answer (1613)

Sol. $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Number of elements is S_1 : $A = A^T \Rightarrow 5^3 \times 5^3$

Number of elements in $A = -A^T \Rightarrow 0$

Similarly, number of elements in $S_3 \Rightarrow$

$$\left.\begin{array}{lll} a_{11}+a_{22}+a_{33}=0 & \Rightarrow & (1,2,-3) \Rightarrow 31 \\ \text{or } (1,1,-2) & \Rightarrow & 3 \\ \text{or } (-1,-1,2) & \Rightarrow & 3 \end{array}\right\} = 12 \times 5^6$$

$n(S_1 \cap S_3) = 12 \times 5^3$

$n(S_1 \cup S_2 \cup S_3) = 5^6(1+12) - 12 \times 5^3$

$\Rightarrow \quad 5^3 \times [13 \times 5^3 - 12] = 125\alpha$

$\alpha = 1613$

23. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{d} = \vec{a} \times \vec{b}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = |\vec{c}|, |\vec{c} - 2\vec{a}|^2 = 8$ and the angle between $\vec{d}$ and $\vec{c}$ is $\frac{\pi}{4}$, then $\left|10 - 3\vec{b} \cdot \vec{c}\right| + \left|\vec{d} \times \vec{c}\right|^2$ is equal to _____.

Answer (6)

Sol. $\vec{a} = \hat{i} + \hat{j} + \hat{k}$

$\vec{b} = 2\hat{i} + 2\hat{j} + \hat{k}$

$\vec{d} = \vec{a} \times \vec{b}$

$= -\hat{i} + \hat{j}$

$|\vec{c} - 2\vec{a}|^2 = 8$

$|\vec{c}|^2 + 4|\vec{a}|^2 - 4\vec{a} \cdot \vec{c} = 8$

$|\vec{c}|^2 + 12 - 4|\vec{c}| = 8$

$|\vec{c}|^2 - 4|\vec{c}| + 4 = 0$

$|\vec{c}|^2 = 2$

$\vec{d} = \vec{a} \times \vec{b}$

$\vec{d} \times \vec{c} = (\vec{a} \times \vec{b}) \times \vec{c}$

$\left(|\vec{d}| \times |\vec{c}| \sin\frac{\pi}{4}\right)^2 = \left((\vec{a}.\vec{c})\vec{b} - (\vec{b}.\vec{c})\vec{a}\right)^2$

$4 = 4|\vec{b}|^2 + (\vec{b}.\vec{c})\,2(|\vec{a}|^2) - 2(\vec{b}.\vec{c})(\vec{a}.\vec{b})$

Let $\vec{b}.\vec{c} = x$

$4 = 36 + 3x^2 - 20x$

$3x^2 - 20x + 32 = 0$

$x = \frac{8}{3}, 4$

$\Rightarrow \quad \vec{b}.\vec{c} = \frac{8}{3}, 4$

$\Rightarrow \quad \vec{b}.\vec{c} = \frac{8}{3}$

Now, $|10 - 3\vec{b}.\vec{c}| + |\vec{d} \times \vec{c}|^2$

$= |10 - 8| + (2)^2 = 6$

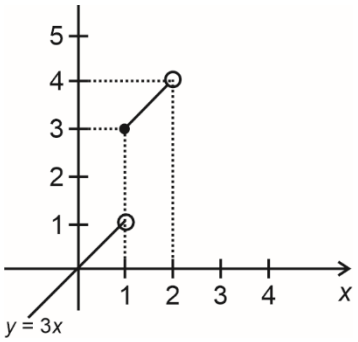


24. Let $f(x)=\begin{cases}3x & ,x<0\\ \min\{1+x[x],\,x+2[x]\} & ,0\le x\le 2\\ 5 & ,x>2\end{cases}$

where [.] denotes greatest integer function. If α and β are the number of points, where f is not continuous and is not differentiable, respectively, then $\alpha+\beta$ equals _____.

Answer (5)

Sol. $f(x)=\begin{cases}3x & ;x<0\\ \min\{1+x,x\} & ;0\le x<1\\ \min\{2+x,x+2\} & ;1\le x<2\\ 5 & ;x>2\end{cases}$



$$f(x)=\begin{cases}3x & ;x<0\\ x & ;0\le x<1\\ x+2 & ;1\le x<2\\ 5 & ;x>2\end{cases}$$

Not continuous at $x\in\{1,2\}\Rightarrow\alpha=2$

Not differentiable at $x\in\{0,1,2\}\Rightarrow\beta=3$

$\therefore\ \alpha+\beta=5$

25. Let $E_1:\frac{x^2}{9}+\frac{y^2}{4}=1$ be an ellipse. Ellipses E_i's is constructed such that their centres and eccentricities are same as that of E_1, and length of minor axis of E_i is the length of major axis of $E_{i+1}(i\ge 1)$. If A_i is the area of the ellipse E_i, then $\frac{5}{\pi}\left(\sum_{i=1}^{\infty}A_i\right)$, is equal to _____.

Answer (54)

Sol. $E_i:\frac{x^2}{9}+\frac{y^2}{4}=1$

Let b_i be minor axis of E_i and a_i be major axis of E_i

$\Rightarrow\ e_i=1-\frac{b_i^2}{a_i^2}$

Now, b_{i+1} be minor axis of E_{i+1} and a_{i+1} be major axis of E_{i+1}

$\Rightarrow\ e_{i+1}=1-\frac{b_{i+1}^2}{a_{i+1}^2}$

Also $a_{i+1}=b_i$ and $e_i=e_{i+1}$

$\Rightarrow\ \frac{b_i^2}{a_i^2}=\frac{b_{i+1}^2}{b_i^2}$

$\Rightarrow\ b_{i+1}=\frac{b_i^2}{a_i}$

$\Rightarrow$ Area of $E_i=S_i=\pi a_ib_i$

$\Rightarrow\ S_{i+1}=\pi a_{i+1}b_{i+1}$

$=\pi(b_i)\left(\frac{b_i^2}{a_i}\right)$

$=\pi(b_ia_i)\left(\frac{b_i}{a_i}\right)^2$

$S_{i+1}=S_i(1-e_i^2)$

$S_{i+1}=S_i(1-\left(1-\frac{4}{9}\right))=S_i\left(\frac{4}{9}\right)$

$\Rightarrow\ S_1=6\pi,\ S_2=6\pi\left(\frac{4}{9}\right),\ S_3=6\pi\left(\frac{4}{9}\right)^2$

$\sum_{k=1}^{\infty}S_i=\left(\frac{6\pi}{1-\frac{4}{9}}\right)=\frac{54\pi}{5}$

$\Rightarrow\ \frac{5}{\pi}\sum_{k=1}^{\infty}S_i=\frac{5}{\pi}\cdot\frac{54\pi}{5}=54$



PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. Three infinitely long wires with linear charge density λ are placed along the x-axis, y-axis and z-axis respectively. Which of the following denotes an equipotential surface?

(1) $xy + yz + zx$ = constant

(2) xyz = constant

(3) $(x^2 + y^2)(y^2 + z^2)(z^2 + x^2)$ = constant

(4) $(x + y)(y + z)(z + x)$ = constant

Answer (3)

Sol. Potential due to an infinite wire is $V = 2k\lambda \ln r$, where r is the distance from the wire.

Taking the point in space $P(x, y, z)$

Distance from wire along x-axis is $r_x = \sqrt{y^2 + z^2}$

Distance from wire along y-axis is $r_y = \sqrt{x^2 + z^2}$

Distance from wire along z-axis is $r_z = \sqrt{x^2 + y^2}$

$\Rightarrow$ Potential at P due to wire along x-axis is

$V_x = 2k\lambda \ln r_x$

Potential at P due to wire along y-axis is

$V_y = 2k\lambda \ln r_y$

Potential at P due to wire along z-axis is

$V_z = 2k\lambda \ln r_z$

$\Rightarrow$ Not potential at $P = V = V_x + V_y + V_z$

or $V = 2k\lambda \ln r_x + 2k\lambda \ln r_y + 2k\lambda \ln r_z$

i.e. $V = 2k\lambda \ln(r_x r_y r_z)$

or $V = 2k\lambda \ln\left(\sqrt{y^2 + z^2}\sqrt{z^2 + x^2}\sqrt{x^2 + y^2}\right)$

$= k\lambda \ln(y^2 + z^2)(z^2 + x^2)(x^2 + y^2)$

$\Rightarrow$ For equipotential surface

$(x^2 + y^2)(y^2 + z^2)(z^2 + x^2)$ = constant

27. Choose the correct nuclear process from the below options

[p: proton, n: neutron, e^-: electron, e^+: positron, ν : neutrino, $\bar{\nu}$: antineutrino]

(1) $n \to p + e^+ + \nu$ (2) $n \to p + e^+ + \bar{\nu}$

(3) $n \to p + e^- + \bar{\nu}$ (4) $n \to p + e^- + \nu$

Answer (3)

Sol. For all nuclear processes, charge must be conserved. Also, when a release of an electron (e^-) is always accompanied by a release of an antineutrino ($\bar{\nu}$)

Hence, $n \to p + e^- + \bar{\nu}$ is the correct answer.

28. Consider a long thin conducting wire carrying a uniform current I. A particle having mass "M" and charge "q" is released at a distance "a" from the wire with a speed v_0 along the direction of current in the wire. The particle gets attracted to the wire due to magnetic force. The particle turns round when it is at distance x from the wire. The value of x is

[μ_0 is vacuum permeability]

(1) $a\left[1 - \frac{mv_0}{q\mu_0 I}\right]$ (2) $\frac{a}{2}$

(3) $a\left[1 - \frac{mv_0}{2q\mu_0 I}\right]$ (4) $ae^{\frac{-4\pi mv_0}{q\mu_0 I}}$

Answer (4)

